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BME 60D

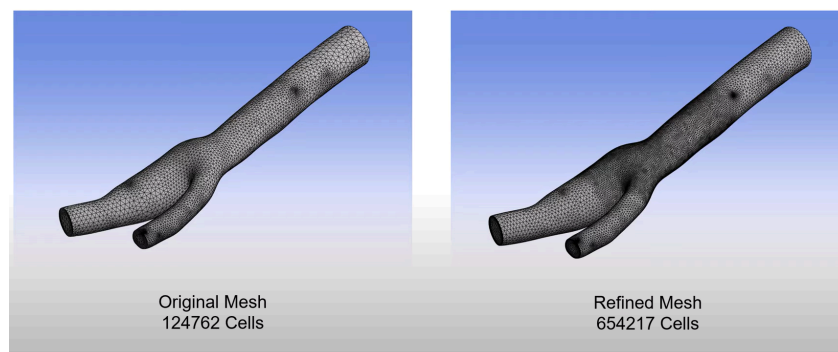
Professor Oomen

### Simulating Blood Flow in a Bifurcating Artery: A CFD Exploration

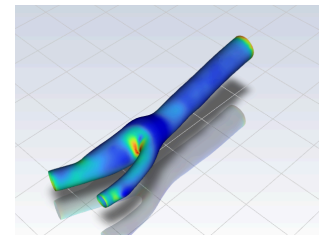
Having worked extensively with solid mechanics simulations, this tutorial provided a compelling contrast through our first hands-on experience with computational fluid dynamics (CFD). Solid simulations revolve around stress, strain, and deformation under applied loads, requiring careful material modeling and boundary condition setups. In contrast, CFD simulations focus on flow behavior, velocity fields, pressure gradients, shear stresses, and mass conservation.

The similarities lie in the mesh generation, boundary condition application, and convergence criteria. Both require verification (e.g., mesh refinement) and validation (e.g., mass flow checks). However, CFD often demands more complex solver setups (turbulence models, multiphase flow options) and is more sensitive to initialization and numerical stability due to the inherently dynamic nature of fluids.

The initial mesh consisted of 124,762 cells, while the refined mesh had 654,217 cells. The refinement was essential for capturing flow details near walls and bifurcations, particularly to resolve wall shear stress gradients. Inaccurate mesh resolution here would lead to erroneous pressure and velocity distributions, especially in curved or branching regions.



In the wall shear stress plot, high stress is observed near the bifurcation apex, highlighting why resolution in this area was necessary. Coarser meshes might smear these gradients, underestimating areas prone to endothelial damage—a key concern in vascular simulations.



Boundary conditions were chosen to emulate arterial blood flow.

- Inlet: A fixed velocity mimicking physiological blood flow ( $\sim 0.6$  m/s).
- 2 Outlets: Set to zero gauge pressure to simulate downstream flow into larger vascular beds.
- Walls: No-slip boundary condition to reflect real vessel walls.

The fluid was modeled as Newtonian, with density  $\approx 1060$  kg/m<sup>3</sup> and viscosity  $\approx 0.0035$  Pa·s. Although blood is non-Newtonian (shear-thinning), at large arteries and high shear rates, the Newtonian assumption is acceptable. For more accurate results in capillaries or under low-shear conditions, a non-Newtonian model (e.g., Carreau-Yasuda) would be more appropriate.

Residuals for velocity components dropped below  $1e-6$ , while continuity dropped to  $\sim 1e-4$ —acceptable for steady-state flow. The drag coefficient ( $C_d$ ) plot stabilized quickly, confirming convergence. The net mass imbalance was  $\sim -5.89e-11$  kg/s, essentially zero, verifying conservation.

As expected, velocity increased along the artery's centerline due to the no-slip condition at the walls. Near the bifurcation, the velocity splits, with outlet 1 receiving 70.2% and outlet 2 receiving 29.8% of the flow. This asymmetry could stem from the geometry or downstream resistance differences. Velocity magnitudes drop near walls and increase near the core, forming parabolic or skewed profiles—classic for laminar, incompressible flow.

Backflow appears slightly in the larger branch after bifurcation. This can be attributed to:

- A sharp bifurcation angle causes flow separation.
- Recirculation zones near the inner walls.
- Pressure build-up due to inertia mismatch at the junction.

These flow phenomena are typical in arterial branches and can influence regions of low wall shear stress, potentially contributing to atherosclerotic plaque development.